Congruence Rigidity and the Fusion Bound: what a positive weight can and cannot do to the spectrum of a transfer kernel

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Abstract

A positive site weight acts on a transfer kernel by *congruence*, $K \mapsto DKD$ with D positive diagonal, not by similarity. Similarity preserves the entire spectrum; congruence preserves strictly less, and strictly more than nothing. We determine both halves for the kernel of L decoupled Ising bonds. *Rigid half:* the sign of every quadratic form value survives, so definiteness in either sign is a congruence invariant — the definite case of Sylvester's law of inertia, recorded here in the elementary form that a machine can check without an eigenvalue count. *Fragile half:* the value of the subdominant ratio r — second largest eigenvalue modulus over the Perron root — does not survive at all, and we locate exactly how far it moves. Restricting the L-site kernel ($L \geq 1$) to the two antipodal configurations leaves *one* Ising bond of coupling βL; for $\beta > 0$ its ratio is $\tanh(\beta L)$. A weight concentrating on the ferromagnetic configurations therefore *fuses* the L sites into a single effective site carrying L times the coupling. Since $\tanh(\beta L) \to 1$ for every fixed $\beta > 0$, there is no bound $r \leq \rho < 1$ holding *simultaneously* in L and over the whole positive-diagonal congruence orbit — which is not the same as ruling one out for a particular physical family of weights. The obstruction to volume-uniformity is named, exactly, and it is a property of the congruence and not of any spectral estimate. Consequently an argument that bounds r using only congruence invariants — rank, inertia, definiteness — cannot produce an L-uniform bound at all. We then ask which congruence is extremal. For *uniform* concentrations the answer is proved: keeping m sites of mutual correlation μ leaves $\mu J + (1-\mu)I$, of ratio $(1-\mu)/(1+(m-1)\mu)$, which is strictly decreasing in m — the more sites the weight keeps, the smaller the ratio it can reach, so the best uniform concentration is a *pair.* For *arbitrary* weights we then prove $\sup_{D>0} r(DMD) = (1-\mu)/(1+\mu)$ with $\mu = \min_{i \neq j} M_{ij}$: the *least correlated pair* determines the supremum, which is approached by concentrating on such a pair. (We evaluate the supremum; we do not classify its maximisers.) The proof is geometric rather than spectral — Hilbert's projective diameter is itself a congruence invariant, and Birkhoff's contraction theorem converts it into the bound — and it uses no definiteness at all, only positivity of the entries. The hypercube instance is the fusion bound. Every new algebraic lemma is machine-checked in Lean 4; the supremum theorem combines those verified lemmas with two cited classical inputs (the Birkhoff spectral bound and finite-dimensional eigenvalue continuity) and is therefore proved at paper level, not end-to-end in the kernel.

1 The question

Let M be a real symmetric matrix and D an invertible diagonal matrix with positive entries. The map

$$M \longmapsto DMD$$

is a *congruence*. It arises whenever a statistical-mechanical transfer kernel is symmetrised against a site weight: if the unnormalised kernel is $w(\sigma)K(\sigma,\tau)$, the symmetric form with the same spectrum is $\sqrt{w(\sigma)}\,K(\sigma,\tau)\sqrt{w(\tau)}$, that is DKD with $D = \mathrm{diag}(\sqrt{w})$.

Congruence is not similarity. Similarity $M \mapsto SMS^{-1}$ preserves the characteristic polynomial and therefore everything spectral. Congruence preserves the quadratic form only up to a change of variable, and the change of variable is not an isometry. The natural question — and, we will argue, the one that governs whether a spectral estimate can be uniform in the system size — is:

Which spectral data survive a positive-diagonal congruence, and how far can the rest move?

The quantity we track. Throughout, the kernels are entrywise positive, so by Perron–Frobenius the spectral radius $\rho(T)$ is itself a simple eigenvalue. We write

$$r(T) \;:=\; \frac{\max\{\,|\lambda| \;:\; \lambda \in \sigma(T),\ \lambda \neq \rho(T)\,\}}{\rho(T)},$$

the *subdominant ratio.* Since we allow indefinite M, this must be said explicitly: ordering eigenvalues algebraically, or by modulus, or by index gives different conventions once negative eigenvalues are present, and it is the quantity above — second largest *modulus* over the Perron root — that Fact 6.1 controls.

We answer both parts for the kernel of L decoupled Ising bonds. Section 2 gives the rigid half, Section 3 the fragile half, Section 4 the consequence, Section 5 the extremal congruence among uniform concentrations, and Section 6 the supremum over all positive weights — exactly, by way of Hilbert's projective diameter.

What this paper does not do. It does not decide whether any particular coupled transfer operator has a spectral gap uniform in its extension. It *explains* why such a question is hard by exhibiting the exact mechanism that destroys uniformity, and explaining is not resolving. No statement here has a consequence for gauge theory in any dimension, and none is claimed.

2 The rigid half: congruence cannot move a sign

Write $q_M(x) = \sum_{i,j} x_i M_{ij} x_j$ for the quadratic form of M.

Lemma 2.1 (Change of variable)**.** *For every M, every d, and every x,*

$$q_{\mathrm{diag}(d)\,M\,\mathrm{diag}(d)}(x) \;=\; q_M(dx), \qquad (dx)_i := d_i x_i.$$

Proof. $(\mathrm{diag}(d)M\,\mathrm{diag}(d))_{ij} = d_i M_{ij} d_j$; substitute and reassociate. □

Theorem 2.2 (Definiteness is a congruence invariant)**.** *Let $d_i \neq 0$ for every i. Then* $\mathrm{diag}(d)M\,\mathrm{diag}(d)$ *is positive definite if and only if M is, and negative definite if and only if M is.*

Proof. By Lemma 2.1 the two forms have the same value set, because $x \mapsto dx$ is a bijection of $\mathbb{R}^n$ (its inverse is $x \mapsto d^{-1}x$) and carries nonzero vectors to nonzero vectors: if $d_i x_i = 0$ with $d_i \neq 0$ then $x_i = 0$. Apply this in both directions. □

Theorem 2.2 is the definite case of Sylvester's law of inertia [1], which says more: the full signature (n_+, n_0, n_-) is a congruence invariant. We do not reprove the general law. We isolate the definite case because it admits a proof through the form alone — no diagonalisation, no eigenvalue count — and that is what makes it cheap to certify mechanically.

Remark 2.3. Invertibility of D is not decorative. A diagonal with k zero entries drops the rank by k and can turn a definite form into a degenerate one. Every claim below assumes D strictly positive.

3 The fragile half: the fusion identity

Let $s(\sigma_j, \tau_j) = +1$ if $\sigma_j = \tau_j$ and -1 otherwise, and let

$$K_\beta^{(L)}(\sigma, \tau) \;=\; \exp\Big(\beta \sum_{j=1}^{L} s(\sigma_j, \tau_j)\Big), \qquad \sigma, \tau \in \{0,1\}^L,$$

the kernel of L decoupled Ising bonds at coupling β. Write $\sigma^+ = (0, \dots, 0)$ and $\sigma^- = (1, \dots, 1)$ for the two antipodal (ferromagnetic) configurations, and let

$$B(a) \;=\; \begin{pmatrix} e^a & e^{-a} \\ e^{-a} & e^a \end{pmatrix}$$

be the transfer matrix of a *single* Ising bond of coupling a.

Lemma 3.1 (One bond)**.** *Let* $a \geq 0$*. Then* $B(a)$ *has eigenvectors* $(1,1)$ *and* $(1,-1)$ *with eigenvalues* $2\cosh a$ *and* $2\sinh a$*, and its subdominant ratio is*

$$r\big(B(a)\big) \;=\; \frac{2\sinh a}{2\cosh a} \;=\; \tanh a.$$

Proof. Direct: $B(a)(1,1)^\top = (e^a + e^{-a})(1,1)^\top$ and $B(a)(1,-1)^\top = (e^a - e^{-a})(1,-1)^\top$. Here $2\cosh a > 0$ is the Perron root, and $a \geq 0$ makes $2\sinh a \geq 0$, so the other eigenvalue is already its own modulus. $\square$

Remark 3.2 (The sign hypothesis is the content)**.** For $a < 0$ the statement is *false*: $2\sinh a < 0$, so the subdominant modulus is $|2\sinh a|$ and $r(B(a)) = |\tanh a|$. An earlier version of this paper omitted the hypothesis, and the formalisation did not catch it — the Lean lemma proves the *algebraic identity* $2\sinh a/2\cosh a = \tanh a$, which holds for every real a, while the English sentence read that identity as a statement about moduli, which it is not unless $a \geq 0$. A true machine-checked lemma can sit underneath a false sentence. Every subsequent *spectral-ratio* consequence uses $a = \beta L$ with $\beta \geq 0$; the block *identity* immediately below is unconditional in β.

That sentence is worth reading twice. In an earlier version it said "everything below uses $\beta > 0$, so the hypothesis costs nothing here" — and the theorem on the next line was then stated for all $\beta \in \mathbb{R}$. The remark documenting a failure to propagate a hypothesis had itself failed to propagate it. The fix is not to phrase the promise better but to make the two statements structurally different, which is what Theorem 3.3 and Corollary 3.4 now are.

Theorem 3.3 (Fusion identity)**.** *For* every $\beta \in \mathbb{R}$ *and every* $L \geq 1$*, the* 2×2 *principal submatrix of* $K_\beta^{(L)}$ *on the antipodal pair* $\{\sigma^+, \sigma^-\}$ *is exactly one Ising bond of coupling* βL*:*

$$\begin{pmatrix} K_\beta^{(L)}(\sigma^+, \sigma^+) & K_\beta^{(L)}(\sigma^+, \sigma^-) \\ K_\beta^{(L)}(\sigma^-, \sigma^+) & K_\beta^{(L)}(\sigma^-, \sigma^-) \end{pmatrix} \;=\; B(\beta L).$$

Corollary 3.4 (Fusion ratio)**.** *If moreover* $\beta \geq 0$*, that block has subdominant ratio* $\tanh(\beta L)$*. For* $\beta < 0$ *its ratio is* $|\tanh(\beta L)|$*.*

Proof. Lemma 3.1 with $a = \beta L \geq 0$; Remark 3.2 for the other sign. $\square$

The split is deliberate, and it mirrors what is and is not machine-checked. The *matrix identity* of Theorem 3.3 is unconditional and is the Lean theorem `antipodal_block_eq_bond`; the *spectral reading* needs the sign hypothesis and is a paper-level consumer. An earlier version merged them and asserted the ratio for all β — false for $\beta < 0$, where $r(B(\beta L)) = |\tanh(\beta L)| > 0$ while $\tanh(\beta L) < 0$. The hypothesis had been added to Lemma 3.1 one paragraph earlier and simply not carried to its consumer.

Proof. σ^+ agrees with itself at all L sites, so the sum of signs is $+L$ and the entry is $e^{\beta L}$; likewise for σ^-. The two antipodal configurations disagree at all L sites, so the sum is $-L$ and the entry is $e^{-\beta L}$. Compare with $B(\beta L)$ and apply Lemma 3.1. □

Theorem 3.3 is the mechanism this paper is about, and it is worth stating in words — carefully, because the natural phrasing overstates. In the singular concentration limit a weight supported on a set S *restricts* the kernel to S; at every strictly positive ε the principal block on S is already preserved *exactly*, and it is the rest of the matrix that shrinks (Proposition 4.3). So the weight does not perturb the block and then correct it; the block is there from the start. When S is the antipodal pair that block is a single bond whose coupling is L times the original. **The weight fuses the L sites into one effective site.**

4 Why that is a wall

Theorem 4.1 (No uniform ratio bound)**.** *For every $\beta > 0$ and every $\rho < 1$ there is an L with $\tanh(\beta L) > \rho$. Consequently the fused ratio is not bounded away from 1, and no estimate of the form $r \leq \rho < 1$ can hold uniformly in the number of sites.*

Proof. $\tanh a = 1 - 2/(e^{2a} + 1)$ is strictly increasing with limit 1; choose L with $e^{2\beta L} + 1 > 2/(1 - \rho)$, which exists because $\beta > 0$ and $e^x > x$. □

This is the honest content of the paper's title. It is *not* a proof that any given coupled transfer operator fails to have a uniform gap; the concentration that realises the fusion is a limit, and the weight of a physical model may or may not approach it. What Theorem 4.1 establishes is that *the obstruction lives in the congruence*, not in whatever spectral estimate one happens to be attempting. An argument that bounds r using only data preserved by congruence — rank, inertia, definiteness — cannot possibly produce an L-uniform bound, because those data are identical for the unfused kernel (ratio $\tanh \beta$) and for every fused one (ratio up to $\tanh(\beta L)$).

Theorem 4.1 is a statement about the scalar $\tanh(\beta L)$; the consequence for the congruence orbit needs one more step, which we make explicit rather than leave to the reader to assemble. Combining Proposition 4.3 with finite-dimensional spectral continuity: for that L and all sufficiently small $\varepsilon > 0$,

$$r\big(D_\varepsilon K_\beta^{(L)} D_\varepsilon\big) \;>\; \rho,$$

with D_ε strictly positive throughout. So the failure of uniformity is realised inside the orbit, not only in its closure.

Corollary 4.2. *Any bound on r that is uniform over both the full positive-diagonal congruence orbit and the number of sites must use a quantity that a positive-diagonal congruence can change.*

The qualification matters. Nothing here forbids a uniform bound for a *particular* physical family of weights; what is forbidden is a uniform bound derived from congruence invariants alone, since those are blind to the difference between the unfused kernel and every fused one.

The orbit, not its closure

The fusion of Theorem 3.3 is realised by a weight that *concentrates*, and the congruences considered here are by *strictly positive invertible* diagonals. Setting the remaining diagonal entries to zero would leave the orbit and land in its closure, so the interface deserves to be stated rather than assumed. It is stated, and the statement is stronger than a limiting argument.

Proposition 4.3 (Concentration stays inside the orbit)**.** *Fix two configurations $p \neq q$ and $0 < \varepsilon \leq 1$, and let D_ε be the diagonal that is 1 at p and q and ε elsewhere. Then D_ε is strictly positive, and for* every *such ε,*

$$(D_\varepsilon M D_\varepsilon)_{pq} = M_{pq}, \qquad (D_\varepsilon M D_\varepsilon)_{qp} = M_{qp},$$

while $|(D_\varepsilon M D_\varepsilon)_{ij}| \leq \varepsilon\, |M_{ij}|$ whenever $i \notin \{p, q\}$.

Proof. $(D_\varepsilon M D_\varepsilon)_{ij} = d_i d_j M_{ij}$ with $d_p = d_q = 1$, giving the two equalities; and $d_i = \varepsilon$, $d_j \leq 1$ off the pair. □

So the fused bond is *already present at every strictly positive ε*: the limit does not create the block, it deletes the rest of the matrix. Converting that deletion into a statement about eigenvalues is the classical continuity of the spectrum in finite dimensions (eigenvalues depend continuously on the entries, e.g. [7, Appendix D]); we cite that step rather than reprove it, and Proposition 4.3 is where the citation attaches.

5 Which congruence is extremal: pairs beat exchangeable clusters

Theorem 3.3 identifies the pair value $\tanh(\beta L)$, approached by concentrating on the antipodal pair — for $L > 1$ it is a statement about a principal *block*, and the orbit of strictly positive congruences reaches $B(\beta L) \oplus 0$ only as $\varepsilon \to 0$ (Proposition 4.3). Is that value the best available? For the family of *uniform* concentrations the question has a complete and elementary answer, and the answer explains the mechanism.

Theorem 5.1 (Exchangeable concentration)**.** *Fix $0 < \mu < 1$ and $m \geq 2$, and let $E_m = \mu J_m + (1-\mu) I_m$ be the exchangeable correlation matrix on m sites. Its spectrum consists of exactly two points: $1 + (m-1)\mu$ on the constant vector, and $1 - \mu$ on every vector summing to zero. Hence the ratio it reaches is*

$$\frac{1-\mu}{1+(m-1)\mu},$$

which is strictly decreasing *in m.*

Proof. $E_m \mathbf{1} = (1 + (m-1)\mu)\mathbf{1}$ by summing a row. If $\sum_i x_i = 0$ then $(E_m x)_i = \mu \sum_j x_j + (1-\mu)x_i = (1-\mu)x_i$. These two eigenspaces have dimensions 1 and $m-1$, so they exhaust the spectrum. For monotonicity, $k \mapsto (1-\mu)/(1+k\mu)$ is strictly decreasing on $k \geq 0$ because the numerator is positive and the denominator strictly increasing. □

The reading is the point. **The more sites an exchangeable concentration keeps, the smaller the ratio it can reach.** So among these the best is the smallest nondegenerate one — a *pair* — and at $m = 2$ the value is $(1-\mu)/(1+\mu)$. On the hypercube kernel, $\mu = e^{-2\beta L}$ at the antipodal pair, and $(1-\mu)/(1+\mu) = \tanh(\beta L)$: Theorem 3.3 is the $m = 2$ case of Theorem 5.1.

Remark 5.2 (Exactly how far this goes)**.** Theorem 5.1 concerns subsets on which *all* pairwise correlations equal a common μ. On a general subset the entries M_{ij} differ and the spectrum has no such closed form, so the theorem does *not* say that a pair beats every uniformly weighted subset of an arbitrary M. The section title should be read with the word *exchangeable* attached. What the theorem does establish is the mechanism — spreading a concentration over more equally correlated sites lowers the reachable ratio — and it is that mechanism, not the general proof; the general proof is §6.

6 The supremum, exactly: Hilbert's projective diameter

The remaining question is what an *arbitrary* positive weight can reach. It has a complete answer, and the answer comes from geometry rather than from spectral estimates: a positive weight cannot change the projective geometry it acts on.

For an entrywise positive matrix T, Hilbert's projective diameter is

$$\Delta(T) \;=\; \max_{i,j,k,l} \log \frac{T_{ik}T_{jl}}{T_{jk}T_{il}}.$$

Birkhoff's theorem [8] says the map induced by T contracts Hilbert's projective metric with coefficient $\tanh(\Delta(T)/4)$, and the spectral consequence is classical.

Fact 6.1 (Birkhoff bound; [8, 9])**.** *For an entrywise positive matrix T with Perron root $\rho(T)$,*

$$r(T) \;\leq\; \tanh\bigl(\Delta(T)/4\bigr).$$

This is the only non-elementary input imported by the upper bound, and it is imported as a statement, not reproved. It is one of *two* classical inputs in this section: the lower bound additionally uses the finite-dimensional continuity of eigenvalues. The two facts that make Fact 6.1 bite are elementary, and both are proved here.

Remark 6.2 (An independent check on the value)**.** The contraction coefficient is also written without logarithms or hyperbolic functions [6]: with $\varphi(A) = \min_{i,j,k,l} A_{ik}A_{jl}/(A_{jk}A_{il})$,

$$\tanh\bigl(\Delta(A)/4\bigr) \;=\; \frac{1-\sqrt{\varphi(A)}}{1+\sqrt{\varphi(A)}}.$$

For our M the cross-ratios come in reciprocal pairs, so $\varphi(M) = 1/e^{\Delta(M)} = \mu^2$ and the right-hand side is $(1-\mu)/(1+\mu)$ directly. The two closed forms agree, which is a useful check: the *upper-bound constant* of Theorem 6.6 can be read off the minimum cross-ratio with no transcendental function anywhere. It is a second derivation of the Birkhoff *coefficient*, not a second proof of the theorem — the equality of the supremum still needs the lower bound, hence the concentration and the spectral continuity.

Lemma 6.3 (Δ is a congruence invariant)**.** *For every positive diagonal D and all indices,*

$$\frac{(DMD)_{ik}(DMD)_{jl}}{(DMD)_{jk}(DMD)_{il}} \;=\; \frac{M_{ik}M_{jl}}{M_{jk}M_{il}}, \qquad \textit{hence} \qquad \Delta(DMD) = \Delta(M).$$

Proof. $(DMD)_{ab} = d_a d_b M_{ab}$, so numerator and denominator both acquire the factor $d_i d_j d_k d_l$, which cancels. □

Lemma 6.4 (The diameter of a unit-diagonal kernel)**.** *Let M be a symmetric $n \times n$ matrix with $n \geq 2$, $M_{ii} = 1$, $M_{ij} \leq 1$, and*

$$\mu \;:=\; \min_{i \neq j} M_{ij} \;>\; 0.$$

Then $\Delta(M) = 2\log(1/\mu)$ exactly.

Proof. Every cross-ratio is at most $1 \cdot 1/(\mu \cdot \mu) = 1/\mu^2$. Taking $i = k$ and $j = l$ on a pair attaining the minimum gives $M_{ii}M_{jj}/(M_{ji}M_{ij}) = 1/\mu^2$, so the maximum is attained. $\square$

Remark 6.5 (μ must be the attained minimum)**.** An earlier version stated this lemma with μ merely a *lower bound*, $\mu \le M_{ij} \le 1$. That is false: $M = J_n$ with $\mu = \frac{1}{2}$ satisfies every such hypothesis, yet all cross-ratios equal 1, so $\Delta(J_n) = 0$ while $2\log(1/\mu) = 2\log 2 > 0$. The proof always used attainment; only the statement did not.

The Lean module had it right and the prose did not, which is worth recording rather than quietly repairing. There, `crossRatio_le_of_bounds` assumes only $\mu \le M_{ab}$ and concludes an *inequality*, while `crossRatio_attained` takes $M_{ij} = \mu$ as an explicit hypothesis. The two halves were kept apart in the kernel and merged in the English.

Theorem 6.6 (Least-correlated pair)**.** *Let M be a symmetric $n \times n$ matrix with $n \ge 2$, $M_{ii} = 1$ and $0 < \mu \le M_{ij} \le 1$ for $i \ne j$, where $\mu = \min_{i \ne j} M_{ij}$. Then*

$$\sup_{D>0 \text{ diagonal}} r(DMD) \;=\; \frac{1-\mu}{1+\mu}.$$

Proof. Upper bound. DMD is entrywise positive, so by Lemmas 6.3 and 6.4 and Fact 6.1,

$$r(DMD) \;\le\; \tanh\bigl(\Delta(DMD)/4\bigr) \;=\; \tanh\bigl(\tfrac{1}{2}\log(1/\mu)\bigr) \;=\; \frac{1/\mu - 1}{1/\mu + 1} \;=\; \frac{1-\mu}{1+\mu},$$

the middle identity because $t = e^{\frac{1}{2}\log(1/\mu)}$ satisfies $t^2 = 1/\mu$ and $\tanh\log t = (t^2-1)/(t^2+1)$.

Lower bound. Fix an argmin pair $\{p, q\}$ and let D_ε be as in Proposition 4.3. Entrywise,

$$T_\varepsilon := D_\varepsilon M D_\varepsilon \xrightarrow[\varepsilon \to 0]{} T_0 = \begin{pmatrix} 1 & \mu \\ \mu & 1 \end{pmatrix} \oplus 0_{n-2},$$

since the block is exactly preserved and every other entry is $O(\varepsilon)$. The spectrum of T_0 is $\{1+\mu,\ 1-\mu,\ 0, \dots, 0\}$, whose Perron root is $1+\mu$ and whose next largest modulus is $1-\mu$. Eigenvalues of a symmetric matrix depend continuously on its entries, so $r(T_\varepsilon) \to (1-\mu)/(1+\mu)$, and the supremum is at least that. $\square$

Remark 6.7 (Attainment, and a retracted claim)**.** An earlier version of this paper asserted that the supremum is *not* attained. That is false. For $n = 2$ the matrix $\left(\begin{smallmatrix} 1 & \mu \\ \mu & 1 \end{smallmatrix}\right)$ is its own least correlated pair and already has eigenvalues $1 \pm \mu$ at $D = I$, so the value is reached with no concentration and no limit; for $\mu = 1$ the matrix is J, the ratio is 0, and the value is reached for every D. The stated reason was worse than unsupported: it claimed Δ is "a strict maximum only in the limit", which Lemma 6.3 directly contradicts — that lemma proves Δ is *constant* on the orbit. **We do not classify attainment here.** Doing so for $n > 2$ would need the equality cases of the Birkhoff bound, which the invariance of Δ alone cannot supply.

Three things about this proof are worth saying plainly.

It uses no definiteness whatever — only positivity of the entries. That is why the twelve indefinite witnesses in the certificate below all obeyed the bound: the pre-registered gate that predicted otherwise was pointing at this argument, and we did not see it at the time.

It explains the rigidity. The quantity that survives the weight is not a spectral one at all; it is the projective diameter. Congruence moves the spectrum but cannot move the geometry, and the reachable spectral ratio is pinned by the geometry.

Novelty, stated carefully. Both imported facts are classical, and Lemma 6.3 is immediate once written down. What we did not find in the searches we ran is this exact *evaluation* of the supremum of the spectral ratio over a diagonal congruence orbit. That is an absence of found prior art, not a claim of priority.

What the first attempt could not do

We record the route that failed, because it locates the difficulty. Since T is positive definite exactly when M is (Theorem 2.2),

$$r(T) \leq \frac{1-\mu}{1+\mu} \iff \lambda_0 - \lambda_1 \ \geq \ \mu\,(\lambda_0 + \lambda_1),$$

and writing $M = \mu J + (1-\mu)N$ — where N again has unit diagonal and entries in $[0,1)$ — gives $T = \mu\, dd^\top + (1-\mu)\, DND$, a rank-one positive semidefinite piece plus an entrywise nonnegative one. Weyl's inequality and the Perron root of the second piece bound $|\lambda_1|$ correctly but lose too much on λ_0: testing against the Perron vector of DND reduces the claim to $\sum_{i\neq j}(1+\mu-2M_{ij})u_i u_j \geq (1-\mu)\sum_i u_i^2$, whose left-hand side is negative once some $M_{ij} > (1+\mu)/2$. The top eigenvector of T is not the Perron vector of DND, and the slack is exactly that gap.

On the hypercube, $K_\beta^{(L)}$ normalised to unit diagonal has $M_{\sigma\tau} = e^{-2\beta h(\sigma,\tau)}$ with h the Hamming distance; the least correlated pair is the antipodal one, $\mu = e^{-2\beta L}$, and $(1-\mu)/(1+\mu) = \tanh(\beta L)$. Theorem 6.6 therefore contains the *spectral-ratio consequence* of Theorem 3.3 (Corollary 3.4) as its hypercube instance; the identity of Theorem 3.3 itself supplies the explicit antipodal block that the lower bound concentrates on.

Numerical certificate. Random search plus Nelder–Mead ascent over positive diagonals, on families chosen to break it and run before the proof was found, never exceeded $(1-\mu)/(1+\mu)$ and in almost every case *matched* it to machine precision. We say "matched" rather than "attained": a floating-point equality at the end of an ascent is not evidence that a maximiser exists, and this paper does not classify maximisers.

family	worst excess over target	cases
hypercube, $L \in \{2,3,4\}$, $\beta \in \{0.15, 0.4, 0.8\}$	1.0×10^{-15}	9
random positive Gram, $n \in \{4,5,6\}$	5.0×10^{-16}	12
full rank / near singular ($\kappa \sim 6 \times 10^6$)	3.3×10^{-16}	4
near-exchangeable, isolated argmin, entries near 1	4.4×10^{-16}	3
exactly exchangeable $\mu J + (1-\mu)I$	1.0×10^{-15}	2
indefinite, $\lambda_{\min} \in [-0.42, -0.06]$	1.6×10^{-15}	12
larger sizes, $n \in \{8,10,12\}$, κ up to 5×10^4	4.2×10^{-16}	6

The last row was added after an audit of our own procedure: the adversarial sweep as first written never actually reached $n = 8$. Its positive-definiteness *precondition* fired on that family, the generator was repaired, and the family was not re-run — so the largest size with evidence had been $n = 6$, not $n = 12$ as the sweep's summary implied. A precondition that is never met carries no evidence either way, and the script now returns a distinct INCONCLUSIVE code rather than reporting such a case as a result.

The last row deserves comment, because it corrected us. A pre-registered gate predicted that positive definiteness would be load-bearing — that at least one of twelve indefinite witnesses would break the bound. None did; all twelve landed *on* it. The hypothesis was therefore dropped, and Theorem 6.6 as stated above assumes no definiteness. The gate, its prediction, and its failure are recorded in the repository rather than repaired in place.

Scope of the certificate. These are numerics, not proof. They cover $n \leq 12$ and $\mu > 0$, and they include indefinite symmetric matrices — but entrywise positivity still guarantees Perron dominance, so the Perron root is the spectral radius in every case and r always means what §1 says it means. What is *not* tested is $\mu \leq 0$, or matrices with nonpositive entries, where both the Birkhoff argument and the definition of the class break down. (An earlier version claimed the sweep did not reach "a negative eigenvalue large enough to make $|\lambda_0|$ the negative extreme". That situation cannot arise here at all: entrywise positivity forbids it. The sentence was a leftover from before r was defined through the Perron root.)

7 Relation to prior work

Sylvester (1852). The law of inertia is classical and Theorem 2.2 is its definite case. We claim no novelty on the rigid half; the contribution is that this form of it is checkable without diagonalisation.

Ostrowski. Ostrowski's quantitative form of the inertia law gives, for Hermitian A and invertible S, $\lambda_k(SAS^*) = \theta_k \lambda_k(A)$ with $\theta_k \in [\lambda_{\min}(SS^*), \lambda_{\max}(SS^*)]$. For $S = D$ diagonal this yields

$$r(DAD) \;\leq\; \kappa(D^2)\, r(A), \qquad \kappa(D^2) = \frac{\max_i d_i^2}{\min_i d_i^2},$$

which **diverges** exactly in the regime of interest, namely when D degenerates. Theorem 6.6 supplies a *finite* substitute in that regime — the exact supremum $(1-\mu)/(1+\mu)$, and $\tanh(\beta L)$ in the hypercube case, however badly conditioned D becomes — while Theorem 3.3 identifies the extremal antipodal block that realises it there. That contrast, divergent classical bound against finite exact value, is stated side by side so that it can be checked.

Onsager and Kaufman. The kernel $DK_\beta D$ with a ferromagnetic nearest-neighbour weight is the transfer matrix of the two-dimensional anisotropic Ising model on a strip, whose spectrum is known in closed form. Nothing in Sections 3–4 is new mathematics about that model, and the reader who knows the exact solution will find $\tanh(\beta L)$ unsurprising: it is the zero-temperature limit in the transverse direction, where the strip collapses to a single spin. The contribution is not the value. It is that the value is *exactly* the supremum over the whole diagonal congruence orbit (Theorem 6.6), that the orbit's rigid invariants are simultaneously pinned, and that the pair of facts identifies which quantities an argument may and may not use. An earlier version of this paper could only conjecture the first of those; the phrasing here is now licensed by §6.

Two branches we have not yet audited, named as obligations. *(i) Diagonal scaling.* Eveson and Nussbaum [9] explicitly connect the Birkhoff–Hopf circle to *DAD theorems*, and there is a literature on $A \mapsto D_1AD_2$ — matrix balancing, Sinkhorn normal form, van der Sluis's optimal scaling [10], surveyed in [5]. The survey's stated scope is the *normalisation* problem: existence and uniqueness of scalings making D_1AD_2 doubly stochastic, and the generalisation to positive maps. We found no treatment of spectral-ratio optimisation over a congruence orbit in the parts we were able to read. **That is a partial reading, not a completed audit**, and we record it at that strength: it is evidence that the branch is aimed elsewhere, not evidence that it lacks the present statement. *(ii) The exact source of Fact 6.1.* Besides [8, 9], the priority list for that hunt is Bushell's projective contraction ratio [4] and the Bauer–Deutsch–Stoer line on bounds for non-dominant eigenvalues of positive operators. Any strengthening of the novelty claim below must go through both branches first.

Formalisation. We are not aware of a mechanised treatment of congruence orbits of transfer kernels. This is an absence of found prior art in the searches we ran, not a claim of priority.

8 What is open

1. Formalising Fact 6.1 and a finite-dimensional eigenvalue-continuity consumer, which would promote Theorem 6.6 from a paper-level proof to an end-to-end Lean theorem. This is the single largest gap between what is written and what is machine-checked. Remark 6.2 suggests the practical route: the φ-form is purely algebraic, so a formalisation need not carry log or tanh at all.

2. Whether the congruence-invariance of Δ extends usefully beyond diagonal congruences — for instance to congruences by positive matrices that are not diagonal, where the cross-ratio no longer cancels.

3. Whether the physical weight w_γ attains the antipodal restriction in the limit $\gamma \to \infty$ with the rate we measure. The deficit $\tanh(\beta L) - r(\gamma)$ was found to decay like $e^{-2\gamma}$ (fitted exponents -1.96 to -2.08 against a pre-registered prediction of -2), consistent with one-domain-wall configurations dominating the correction. The limit itself is not proved here.

4. The general signature-counting law as a mechanised statement.

Machine verification — status

The following are written in Lean 4: the rigid half (Theorem 2.2, Lemma 2.1); the *matrix identity* underlying Theorem 3.3; the *scalar* tanh-*approach lemma* underlying Theorem 4.1; Proposition 4.3; and the algebraic and eigenvector ingredients of Lemma 3.1, Theorem 5.1 and Theorem 6.6 against a pinned Mathlib in `YangMills/OS/CongruenceSpectrum.lean`, and **have been verified**, including the two lemmas that carry Theorem 6.6 and the $n = 2$ attainment witness of Remark 6.7.

What "ingredients" means, precisely. The module contains no Lean definition of r, no enumeration of a spectrum, and no theorem asserting that a second largest *modulus* equals any stated quantity. What it contains for Lemma 3.1 and Theorem 5.1 is the eigenvector equations, the algebraic ratio identity, the positivity and sign lemmas, and the scalar monotonicity. The same caution applies to Theorems 3.3 and 4.1: Lean proves the antipodal block equals $B(\beta L)$ as matrices, and proves $\forall \rho < 1\ \exists L,\ \rho < \tanh(\beta L)$ as a scalar statement, but the reading of either as a claim about the spectrally defined r — and the passage from the block to a strictly positive member of the orbit — is completed at paper level. *Their interpretation in terms of the spectrally defined r of §1 is completed at paper level*, by elementary finite-dimensional linear algebra. We are explicit about this because it is exactly the gap Remark 3.2 is about: a true Lean lemma can sit underneath a sentence that says more than it does, and this paper has already published one such sentence.

Theorem 6.6 is *not* an end-to-end Lean theorem, and we do not claim it is. It is proved at paper level from two machine-checked new lemmas (Lemmas 6.3 and 6.4, plus the hyperbolic identity) together with two cited classical inputs that are *not* formalised: Fact 6.1, and the finite-dimensional continuity of eigenvalues used in its lower bound. Neither is a Lean theorem in the dependency cone of the 40 endpoints, so those endpoints certify the algebra, the congruence invariance, the exact value of the diameter and the hyperbolic identity — not the imported spectral inequality. Formalising Fact 6.1, or importing it through a clean Lean interface, is the obvious next step and is listed in §8.

One further caveat of provenance. The bibliographic data for [8, 9, 6] were checked against the publishers' records, but the internal numbering of the results inside those sources was not, and *two further attempts to pin it failed*: neither the publicly accessible material for [9] nor the secondary literature we could reach states the spectral corollary with a locatable number. Fact 6.1 is therefore attributed to the works rather than to a numbered statement in them. **Pinning that numbered statement is a prerequisite for journal submission**, not an optional polish: Fact 6.1 is the single *non-elementary* load-bearing import in the *upper bound* of Theorem 6.6 — the lower bound imports finite-dimensional spectral continuity as well — and a reader must be able to go to one line of one source and check it. Remark 6.2 reduces the exposure somewhat by giving a second closed form that reproduces the same value, but it does not remove it.

No statement in this paper is conjectural.

commit	`b36f50d1`
module `sha256`	`56a66ee1062108fc7419da3c` `f7cfcb315a2f5809f20d31d1a3806fcc68334a5e`
toolchain	`leanprover/lean4:v4.29.0-rc6`
Mathlib pin	`07642720480157414db592fa85b626dafb71355b`
elaboration	exit `0`; empty log — zero errors, zero warnings, zero `sorry`
axiom oracle	40/40 declarations on `[propext, Classical.choice, Quot.sound]`
`sorryAx`	none

The module hash was computed independently on the author's machine and on the verification host and agrees, so the checked object is the source printed here. Compilation ran on a CPU (never GPU) high-memory cloud runtime, per the project's environment rule of 1 August 2026; the desktop compiles nothing.

One registered check remains undone. The pre-registration also asked that the module be imported into the project's aggregate library and that the build job count increment by exactly one. It has not been imported, so there is no increment to measure; measuring it honestly requires two full builds of the aggregate in the same tree, which the shared Mathlib cache does not cover. That check is therefore reported as *not run* rather than dropped, and it is not replaced by a comparison against a count measured elsewhere. Nothing in the theorems above depends on it.

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